

**DEVELOPMENT OF A PSYCHOMETRIC-BASED COLLEGE
RECOMMENDATION SYSTEM USING STUDENT PROFILING**



A Capstone Project **PROPOSAL**

Presented to

The Faculty of the College of Information Technology

Tagoloan Community College

Tagoloan, Misamis Oriental

In Partial Fulfillment

Of the Requirements for the Degree

Bachelor of Science in Information Technology

By

COMENDADOR, JOHN WILSON C.

DAYONDON, AIZA MAE P.

GATCHALIAN, RHONALD MARC B.

MAPALO, JORI LLOYD L.

VIRADOR, EMIEL EARL CHRISTIAN B.

MAY 2026

TABLE OF CONTENTS

	Page
TITLE PAGE	i
TABLE OF CONTENTS	ii
CHAPTER	
1 BACKGROUND OF THE STUDY	
Project Context	1
Purpose and Description	2
Conceptual Framework	3
Objectives of the Study	4
Scope and Limitation	5
Definition of Terms	6
2 REVIEW OF RELATED LITERATURE AND STUDIES	8
3 RESEARCH METHODOLOGY	15
Planning	16
Designing	16
Development	17
Testing	18
Deployment	19
Review	19
Launch	20
REFERENCES	21
APPENDICES	25

CHAPTER 1

BACKGROUND OF THE STUDY

Project Context

The pursuit of academic excellence is fundamentally rooted in the synergy between an individual's innate competence and their vocational interests. Ensuring a seamless transition from secondary to tertiary education is a critical requirement for achieving United Nations Sustainable Development Goal 4 (SDG 4): Quality Education (United Nations, 2023). In the Philippine educational context, diagnostic systems such as the National Career Assessment Examination (NCAE) are widely used to assess students' aptitudes and guide them toward appropriate academic tracks and career paths, covering areas such as General Scholastic Aptitude, Scientific Ability, Mathematical Ability, Reading Comprehension, Technical-Vocational Aptitude, and interest inventories based on the RIASEC model (Department of Education, 2025). However, despite the use of these standardized assessments across schools, several real-world challenges persist, including mismatch between students' chosen courses and their assessed abilities, influence of peers or family in decision-making, lack of personalized guidance, and inefficient manual processing of student data, which often lead to low academic performance, decreased motivation, and course shifting (CHED, 2023). In the field of Information Technology, the development of diagnostic systems such as psychometric-based recommendation systems, decision support systems, and student profiling platforms has emerged as a practical solution to these issues, as they utilize standardized assessments like NCAE and institutional admission examinations to generate data-driven insights, minimize human error, and improve the accuracy of student-course matching (Stair & Reynolds, 2020).

Tagoloan Community College (TCC) is a local government-funded higher education institution located in Misamis Oriental, Philippines. As a major academic provider in the region, the college processes approximately 8,000 student applications annually. At present, the institution utilizes manual procedures and established admission examination thresholds to manage its admission process. This locale was selected for the study to observe how admission examination scores and RIASEC assessment results are used to provide students with appropriate course recommendations. By focusing on TCC, the researchers aim to evaluate how personal interest profiles and test performance can be integrated into the existing administrative framework to suggest suitable academic paths for a large applicant pool.

Currently, the institution relies on a singular aggregate admission exam score to determine eligibility for board-regulated courses (requiring a score of 2.50 or better). This manual approach results in an "academic mismatch" for thousands of students who fail to meet specific cut-offs or are undecided about their alternative options. As noted by Pascua and Navalta (2023), students who enter courses without a clear understanding of their vocational fit are significantly more likely to shift or drop out. (Quintos et al., 2022).

performance with interest profiles, the research provides a data-driven basis for course recommendations for all examinees. The primary goal of this work is to support informed

academic decision-making and enable the institution to handle student information more systematically, ensuring that every applicant receives guidance toward a suitable academic path.

Purpose and Description

This study aims to develop a Psychometric-Based College Course Recommendation System Using Student Profiling and RIASEC Psychometric Profiling for Tagoloan Community College. The system is intended to support and improve the admission process by providing data-driven course recommendations based on the student's admission examination results and vocational interests. It utilizes the admission examination results as an indicator of student competence, while a RIASEC-based assessment is used to identify the student's interests and career inclinations through a set of structured questions. By integrating these components, the system will be able to generate course recommendations that are more appropriate for each student and support informed and data-driven decision-making during admission (Su et al., 2021; Hirschi, 2020).

The study will benefit incoming students by providing a structured and data-driven approach in selecting academic courses that align with their competence and interests. Many students experience difficulty in choosing a course due to uncertainty, lack of guidance, or limited awareness of suitable options. Through the system, students are given recommendations that consider both their admission examination results and personal preferences, which can help them better understand their possible career paths. When there is a strong alignment between a student's competence and interests and their chosen course, it can contribute to improved academic satisfaction, motivation, and persistence (Kaya & Karakaya, 2021; Rudolph et al., 2023).

Furthermore, when there is a mismatch between the student's competence and interests and their chosen course, it may increase the likelihood of shifting courses or discontinuing their studies. The system aims to minimize this by providing more accurate and data-driven course recommendations. The admission office of Tagoloan Community College will also benefit from the implementation of the system by enhancing the efficiency and organization of the evaluation and placement process. The current reliance on manual processing can be time-consuming and may lead to inconsistencies, especially when handling a large volume of applicants. With the proposed system, student data can be processed more systematically, and course recommendations can be generated in a consistent and timely manner, resulting in a more streamlined and reliable admission workflow (Borgen & Betz, 2022; Su et al., 2021).

Conceptual Framework

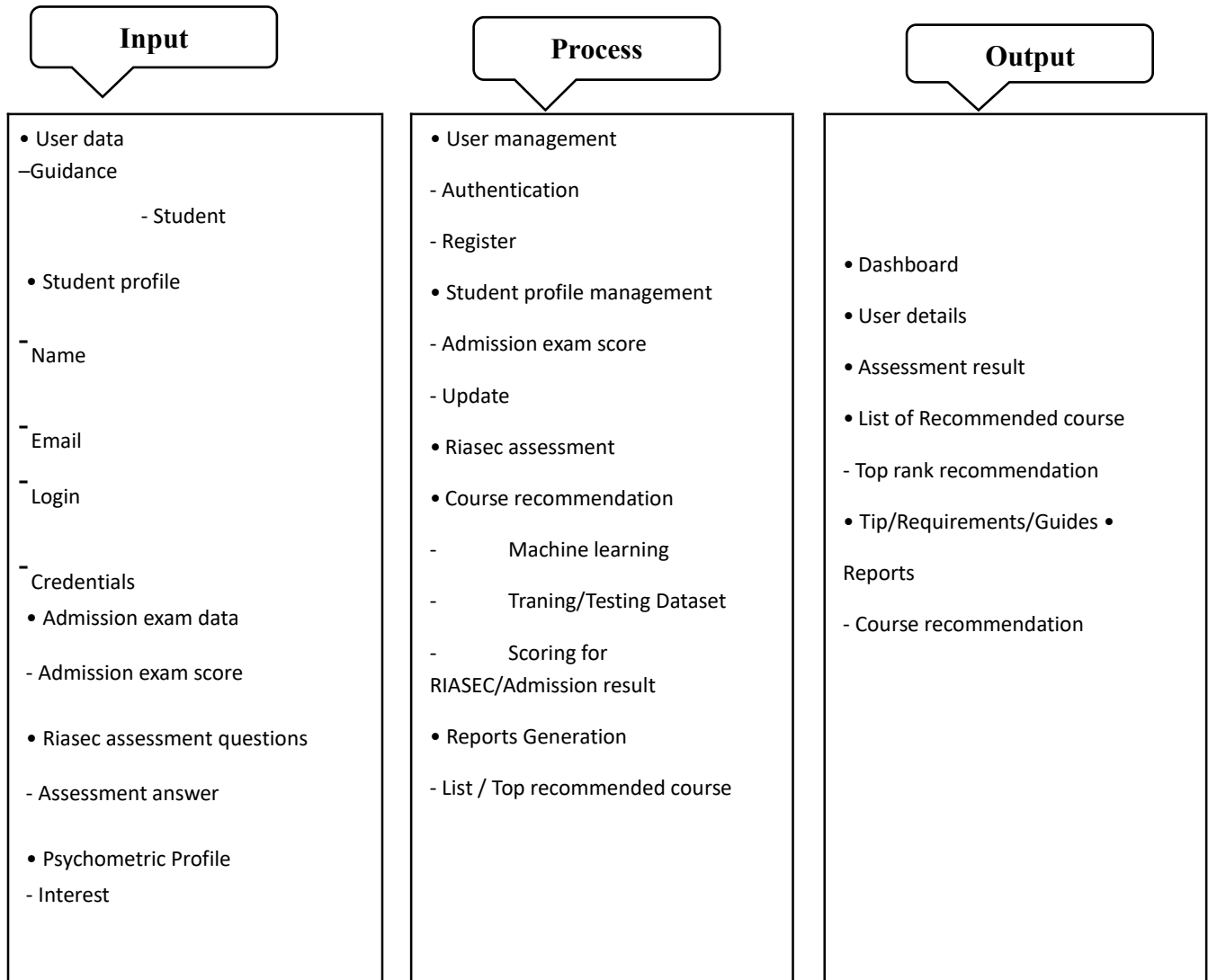


Figure 1.0 Conceptual Framework

The input stage consists of all data collected from the user, including personal information such as name, email, login credentials, and student profile details. It also includes admission exam scores and responses to the RIASEC assessment to identify students' interests and abilities. These inputs serve as the foundation of the system to ensure accurate and relevant recommendations based on the Input model (Xingran Zhao & Supinda Lertlit, 2025).

The process stage involves the system's operations that handle and analyze the input data, including user management functions such as registration, authentication, and profile updates. The system applies RIASEC assessment scoring and uses machine learning techniques to evaluate student profiles. It transforms input data into meaningful results by matching students with suitable college courses based on the IPO framework (Saiqa Bashir et al., 2026).

The output stage presents the processed results in a clear and organized manner through dashboards displaying user details, assessment results, and recommended courses. It highlights top-ranked recommendations and provides additional tips, requirements, or guides. These outputs support informed decision-making and represent the final stage of the IPO framework (Xingran Zhao & Supinda Lertlit, 2025).

Objective of the Study

The general objective of the study is to design and develop a Psychometric-Based College Course Recommendation System Using Student Profiling. To achieve the general objective, the study will pursue the following specific objectives:

The general objective of the study is to design and develop a Psychometric-Based College Course Recommendation System Using Student Profiling. To achieve the general objective, the study will pursue the following specific objectives:

- a) To gather data on students' psychometric assessment, admission exam results, personality, and career interests using student profile information and psychometric tests (Saya et al., 2025).
- b) To design system diagrams such as Use Case Diagram, Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), manual process, flowchart, and multi-tier architecture using Draw.io (Saputra et al., 2026).
- c) To develop the following functions/modules:
 - a) User Management Module: To address unauthorized access and inaccurate data encoding.
 - b) Authentication: Ensuring secure identity verification for users.
 - c) Registration: Providing a systematic way for students to create accounts and encode data.
 - d) Student Profile Management Module: To address inconsistent data handling by allowing for the Update and management of student records and Admission Exam Scores.

- e) RIASEC Assessment Module: To address the lack of structured evaluation by automating the Scoring for RIASEC and Admission results based on student answers.
 - f) Course Recommendation Module: To address the risk of academic mismatch by utilizing Machine Learning (including Training/Testing Datasets) to cross-reference entrance exam performance with vocational interest profiles.
 - g) Reports Generation Module: To address difficulty in manual data analysis by automatically presenting the List of Recommended Programs, Top Rank Recommendations, and a formal Course Recommendation Report.
- d) To deploy the system on a web-based platform using Hostinger, HTML, CSS, JavaScript, Bootstrap, and Laravel (Saputra et al., 2026).
- e) To test the system using scenario-based testing through test case execution.
- f) To evaluate the system using the System Usability Scale (SUS) (Saputra et al., 2026).

Scope and Limitation

This study focuses on the Development of a Psychometric-Based College Course Recommendation System Using Student Profiling for Tagoloan Community College (TCC). The system is designed as a specialized computing tool for the Guidance, Testing, and Admission Office to systematically manage and guide the institution's large applicant pool. The scope of the system encompasses all incoming students who undergo the institutional admission examination. It utilizes two primary data points to generate insights the student's aggregate admission examination score and their vocational interests measured through an integrated RIASEC modelbased assessment. The system is programmed to evaluate the overall exam score against TCC's grading policy specifically the 2.50 threshold to filter eligibility for board-regulated and non-board courses. By cross-referencing these scores with dominant vocational interests, the system provides data-driven recommendations for every examinee, including those who are undecided or fail to meet specific course cut-offs. The study covers the design, development, and testing of the student profiling module, the interest assessment tool, and the recommendation engine.

However, the study is limited to the context of Tagoloan Community College and focuses only on incoming students undergoing the admission process. The recommendations generated are intended to guide decision-making and do not replace the student's final choice. The system's accuracy depends on the reliability of admission examination results and the honesty of responses in the RIASEC assessment; inaccurate inputs may lead to less effective recommendations. Furthermore, the study relies solely on the RIASEC model in measuring vocational interests, which may not fully capture other influencing factors such as personal circumstances (e.g., family responsibilities, health conditions, proximity to school, transportation

availability, and the need to work while studying), financial capability, and external pressures. The recommendations are also limited to the courses currently offered by TCC and follow existing admission policies, without considering external opportunities or future course changes. In addition, the study does not include long-term tracking of student performance; therefore, it cannot fully evaluate the effectiveness of the recommendations in terms of academic success, retention, or graduation outcomes. Lastly, as a decision-support tool, the system allows students to accept or reject recommendations, meaning that course mismatches may still occur due to personal or external influences (Pascua & Navalta, 2023; Quintos et al., 2022; Stair & Reynolds, 2020; DepEd, 2023).

Definition of Terms

Admission Examination. A standardized test used to assess a student's academic ability for college admission. In this study, it refers to the institutional exam score used by TCC to determine course eligibility.

Academic Mismatch. A condition where a student's abilities and interests do not align with their chosen academic course. In this study, it refers to students enrolling in courses not suited to their exam results and RIASEC profile.

Career Recommendation System. A decision-support system that suggests career paths based on user data and algorithms. In this study, it refers to the developed system that suggests college courses based on student profiling.

Data-Driven Decision Making. The process of making decisions based on data analysis rather than intuition. In this study, it refers to using exam scores and RIASEC results to recommend academic courses.

Decision Support System (DSS). A computer-based system that supports decisionmaking activities using data and models. In this study, it refers to the system assisting the admission office in evaluating student data.

Machine Learning. A branch of artificial intelligence that enables systems to learn from data patterns. In this study, it refers to the algorithm used to match student profiles with suitable courses.

RIASEC Model. A theory developed by John Holland that classifies personality into six types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. In this study, it refers to the assessment framework used to determine students' vocational interests.

Student Profiling. The process of collecting and analyzing personal and academic data of students. In this study, it refers to gathering student information such as exam scores and interests for recommendations.

Vocational Interest. An individual's preference for certain types of work or career fields. In this study, it refers to the student's dominant RIASEC category used in course matching.

Course Recommendation. A suggested academic or career path based on evaluation criteria. In this study, it refers to the ranked list of college courses generated by the system.

User Management. The process of handling user access, authentication, and account data in a system. In this study, it refers to system functions like login, registration, and account control.

Dashboard. A graphical interface that displays key information and system outputs. In this study, it refers to the interface showing student results and recommendations.

System Usability Scale (SUS). A standardized tool used to evaluate the usability of a system. In this study, it refers to the method used to assess user satisfaction with the system.

Recommendation Algorithm. A computational method used to generate suggestions based on input data. In this study, it refers to the logic used to match exam scores and RIASEC results.

Training Dataset. A set of data used to train machine learning models. In this study, it refers to sample data used to improve recommendation accuracy.

Testing Dataset. A data set used to evaluate the performance of a trained model. In this study, it refers to data used to validate the system's recommendations.

Information System. An organized system for collecting, processing, and delivering information. In this study, it refers to the developed web-based recommendation platform.

Web-Based System. A system accessed through a web browser over the internet. In this study, it refers to the platform deployed online for student use.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDY

Introduction

This chapter presents a collection of related literature, studies, concepts, and existing systems relevant of the proposed development of a psychometric based college course recommendation system using student profiling to help students identify academic programs that match their skills, interests, personality, and career preferences. These systems are designed to support students in making more informed educational decisions by analyzing factors such as psychometric evaluation results, academic performance, and personal characteristics. The related literature, studies, concepts, and existing systems gathered from local and international sources provide the foundation for understanding how course recommendation technologies are applied in educational settings.

Related Literature

Educational institutions increasingly utilize psychometric career guidance systems to improve the accuracy and reliability of academic advising and student placement processes. These systems employ standardized assessments to evaluate cognitive abilities, vocational interests, and personality traits, allowing institutions to provide more structured and evidence based recommendations for learners. According to Ragunath et al. (2026), psychometric based systems overcome the limitations of traditional counseling approaches by systematically analyzing student aptitude and preference indicators to generate personalized academic guidance. Through data driven evaluation, institutions can minimize subjective decision making and improve the consistency of course recommendation procedures. This concept strongly supports the present study because the proposed system seeks to enhance the institution's current admission and advising process through automated psychometric assessment. By integrating measurable student data into the recommendation process, the system is expected to generate more reliable and appropriate course suggestions for incoming students.

Moreover, various psychometric assessment frameworks have been developed to analyze personality, aptitude, interests, and behavioral tendencies that may influence educational and career decisions. Among the most recognized models are the Myers Briggs Type Indicator (MBTI), the Big Five Personality Model, DISC Assessment, Multiple Intelligence Theory, Emotional Intelligence assessment, and Holland's RIASEC model. Although these frameworks contribute significantly to psychological and behavioral evaluation, many of them primarily focus on personality structure, emotional behavior, communication style, or learning preferences

rather than direct vocational alignment. Gardner and Martinko (1996) explained that MBTI is commonly used to identify personality preferences and behavioral patterns, while Stajkovic et al. (2018) emphasized that the Big Five model is more effective in predicting behavioral consistency and psychological traits than vocational compatibility. DISC Assessment and Emotional Intelligence models are also widely applied in leadership, communication, and workplace interaction studies, whereas Multiple Intelligence Theory focuses on identifying learner strengths and instructional adaptation. Despite their relevance in behavioral analysis, these frameworks are less specialized for educational course recommendation and vocational matching.

Furthermore, among these psychometric approaches, Holland's RIASEC model remains one of the most established and empirically validated frameworks for career guidance and vocational alignment because it directly associates student interests with compatible educational and occupational environments. The framework categorizes individuals into six vocational personality types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional. Holland (1997) emphasized that individuals are more likely to achieve academic satisfaction and career success when their vocational interests align with their chosen educational field or profession. Supporting this perspective, Su, Stoll, and Rounds (2021) found that vocational interest alignment significantly affects educational motivation, academic persistence, and long term career satisfaction. Unlike generalized personality assessments such as MBTI and DISC, the RIASEC model specifically evaluates vocational interests and educational compatibility, making it more appropriate for course recommendation systems and career guidance platforms. This literature strongly supports the present study because the proposed system integrates RIASEC assessment alongside admission examination results to generate personalized and academically aligned course recommendations. Through this approach, the system aims to reduce academic mismatch, course shifting, and poor educational alignment among incoming students in the institution.

In addition, modern educational guidance systems increasingly utilize psychological profiling to create personalized academic recommendations based on learner behavior, interests, cognitive characteristics, and educational preferences. Chen and Wang (2026) argued that traditional counseling methods often lack personalization and adaptability, resulting in less effective educational guidance for diverse student populations. Through automated profiling and data driven assessment, institutions can analyze multiple learner characteristics simultaneously instead of relying solely on academic grades or examination scores. This multidimensional approach enables recommendation systems to identify educational pathways that better reflect student strengths, preferences, and capabilities. The relevance of this literature to the present study is significant because the proposed system incorporates student profiling mechanisms to improve the quality and relevance of course recommendations. By integrating psychological and vocational indicators into the evaluation process, the system addresses the limitations of the institution's current admission procedure, where individualized student characteristics are often overlooked.

Additionally, higher education institutions continue to face challenges associated with academic mismatch, particularly among students who select courses without sufficient career assessment and vocational guidance. Poor educational alignment frequently leads to low

academic performance, reduced motivation, course shifting, and student dropout. According to Pascua and Navalta (2023), students who enter academic programs without proper self assessment and career guidance are more likely to experience dissatisfaction and academic difficulty. Similarly, Quintos and Caballes (2022) explained that weak career alignment mechanisms contribute significantly to the mismatch between student interests and chosen academic pathways. External influences such as peer pressure, family expectations, and limited access to structured counseling further increase the likelihood of inappropriate course selection. These findings strongly justify the necessity of the present study because the institution's existing admission process primarily depends on examination cut off scores and manual advising procedures. Through the integration of psychometric assessment, vocational interest analysis, and student profiling, the proposed system seeks to minimize academic mismatch by recommending courses that align with both student competence and career inclination.

Likewise, the growing reliance on intelligent technologies in education has encouraged institutions to adopt machine learning based recommendation systems to improve academic advising and decision making processes. These technologies utilize algorithms capable of analyzing multidimensional datasets to identify patterns and generate recommendations that would otherwise require extensive manual evaluation. Al Dossari et al. (2024) confirmed that machine learning based recommendation engines consistently improve the precision and consistency of career guidance by processing student records and generating data driven academic suggestions. Predictive analysis and automated pattern recognition also enable institutions to evaluate student compatibility with specific academic programs more efficiently. This literature is highly relevant to the present study because the proposed system applies automated recommendation logic to compare student profiles against program requirements within the institution. As the institution handles a large number of student applicants annually, integrating machine learning concepts into the recommendation process helps ensure consistent, accurate, and efficient course evaluation while reducing the possibility of human error.

Subsequently, the increasing complexity of admission and academic advising procedures has encouraged educational institutions to adopt decision support technologies that improve efficiency, consistency, and data driven evaluation. Ogunniran and Makinde (2022) emphasized that decision making is an essential administrative skill in education, explaining that properly designed Decision Support Systems (DSS) function as interactive, knowledge based software tools that help administrators compile raw data, documents, and institutional knowledge to solve non routine managerial problems. Within educational environments, these tools streamline complex processes like student enrollment, career guidance, and academic tracking because they provide systematic mechanisms for capturing and evaluating large volumes of data across multiple systems. The proposed Psychometric Based College Course Recommendation System functions as an automated decision support tool for the institution by calculating admission examination scores and RIASEC assessment results simultaneously. Through this automated approach, the system aims to strengthen the reliability, consistency, and efficiency of course recommendation procedures, effectively operationalizing data driven advising while reducing dependence on purely manual evaluation.

Meanwhile, the growing demand for accessible and scalable digital services has encouraged educational institutions to implement web based platforms for student management, admission processing, and academic evaluation. Compared to traditional standalone systems, web based technologies provide centralized data management, real time processing, accessibility, and multi user functionality across internet enabled devices. According to Grepon et al. (2021), educational institutions benefit from centralized web based systems because they improve data storage, retrieval, and processing efficiency while reducing paperwork and manual record handling. Similarly, Ugalingan and Antonio (2025) emphasized that web based management systems enhance administrative efficiency, improve record accuracy, and support faster information retrieval through automated and centralized operations. These characteristics strongly support the present study because the proposed system is designed as a web based platform capable of accommodating large numbers of applicants within the institution. Through the use of Laravel and modern web technologies, the system seeks to streamline admission procedures, organize student information systematically, and provide accessible recommendation services for both students and administrators.

Related Studies

In this study, according to Minakshi Roy and Kalpana Sharma (2026), an AI-driven career guidance system was developed that combines the RIASEC model with machine learning techniques to recommend appropriate career paths. The system was capable of producing accurate and personalized suggestions, thereby improving users' decision-making processes. However, the study was limited by its reliance on large volumes of high quality data, which may reduce its effectiveness when user information is incomplete. This limitation highlights a gap addressed in the current study, which aims to develop a psychometric-based course recommendation system that can still deliver reliable and personalized results even with minimal student input.

According to Wen Zhang (2026), a deep learning-based career recommendation system was developed to analyze user data and generate long term career pathways. The system demonstrated the ability to deliver highly accurate, data driven recommendations, enhancing the decision-making process of users. However, the study was constrained by its complex design and high computational demands, making it less practical for smaller institutions with limited resources. This limitation underscores the need for a more accessible approach, which the current study addresses by proposing a simplified psychometric-based course recommendation system that balances accuracy with efficiency and usability.

Moreover, according to Su, Liao, Xu, and Zhao (2021), a personality driven recommender system was developed using Holland Code assessments to guide students in selecting appropriate academic courses and career opportunities. Personality traits, interests, and behavioral analysis were incorporated into the recommendation process to generate personalized suggestions for learners. Students were able to identify courses that matched their personalities and preferences more effectively. However, the study focused mainly on personality assessments

without including aptitude testing or broader psychometric evaluations. Depending only on personality traits may not provide accurate and complete course recommendations for incoming freshmen. The concepts presented support the integration of psychometric profiling in educational recommendation platforms.

Furthermore, according to Li and Kim (2021), a deep learning based course recommender approach was designed to provide intelligent academic recommendations through the analysis of students' learning behaviors and educational data. Artificial intelligence and deep learning techniques were applied to improve recommendation accuracy and support students in choosing suitable educational content. Personalized learning experiences and recommendation efficiency were improved through AI based analysis. However, one limitation was the reliance mainly on academic data while excluding psychological profiling, aptitude assessments, and student interests. Combining academic performance with psychometric and interest based evaluations can produce more effective recommendations for students. The ideas presented contribute to the development of AI integrated educational guidance systems.

Additionally, according to Hyland, Hoff, and Rounds (2022), interest ability profile assessments were used to examine the relationship between students' interests, abilities, and knowledge acquisition. Interest inventories and ability assessments were integrated to identify students' strengths, learning capabilities, and possible career suitability. Combining interests and abilities provided a more comprehensive understanding of students' educational potential. However, the study concentrated mainly on assessment analysis without developing an automated recommendation process capable of generating course suggestions. Intelligent platforms are still needed to automatically interpret assessment results and provide recommendations for students. The information presented strengthens the importance of combining aptitude and interest assessments in educational recommendation systems.

According to Ma, Zhang, and colleagues (2021), an interactive course recommendation platform called Course was developed to assist university students in exploring academic subjects through visual and interactive interfaces. User friendly features were designed to improve students' understanding of recommended courses and support academic planning. Students became more engaged and satisfied when using interactive recommendation platforms. However, the study emphasized interface design and user interaction rather than incorporating psychometric profiling, aptitude testing, and interest assessments into the recommendation process. Recommendation platforms should not only provide interactive features but also generate intelligent suggestions based on students' profiles and abilities. The platform demonstrates the value of improving user experience in educational recommendation environments.

Finally, according to Noor Intan Dyana Azlan, Mohd Khalid Awang, and Abd Rasid Mamat (2017), a course recommendation system was developed to help students select suitable academic programs based on their academic qualifications and personality types. The study utilized a rule-based technique combined with academic examination results and John Holland's RIASEC Theory to generate course recommendations. The results showed that the system was able to provide appropriate course suggestions that matched students' abilities and personality

profiles. However, the study was limited to rule-based recommendations within a specific institution, which may affect its flexibility in broader educational settings. This limitation is relevant to the present study, as the proposed system also focuses on providing structured and reliable course recommendations based on student profiling and psychometric assessment results.

Synthesis of the Study

The gathered literature and studies strongly support the development of the proposed Psychometric-Based College Course Recommendation System Using Student Profiling. The reviewed sources consistently revealed that many students experience difficulty in choosing the right college course due to lack of proper guidance, limited self-assessment, and dependence on external influences such as family expectations, peer pressure, and uncertainty regarding their own abilities and interests. Because of this, many students end up enrolling in courses that do not match their capabilities and vocational interests, resulting in academic mismatch, low academic performance, loss of motivation, course shifting, and student dropout. These findings clearly justify the need for a more reliable, systematic, and data-driven recommendation process that can assist students in making better academic decisions.

The reviewed literature also showed that educational institutions are increasingly adopting psychometric assessment systems, student profiling technologies, machine learning recommendation systems, and decision support systems to improve academic advising and course recommendation processes. These technologies help institutions analyze student information more efficiently and generate recommendations that are more accurate, personalized, and consistent compared to traditional manual evaluation methods. Through automated processing and intelligent recommendation mechanisms, institutions are able to reduce human error, improve efficiency, and provide students with educational pathways that better match their strengths, interests, and academic capabilities.

Furthermore, several psychometric frameworks were discussed in the reviewed literature, including MBTI, Big Five Personality Model, DISC Assessment, Multiple Intelligence Theory, and Holland's RIASEC model. Although these frameworks are useful in identifying personality traits, behavioral patterns, communication styles, and learning preferences, most of them are not specifically designed for direct vocational and educational alignment. Some frameworks mainly focus on personality behavior and emotional characteristics rather than determining whether a student's interests fit a particular academic field or career environment.

Among these psychometric approaches, the RIASEC model was identified as the most appropriate framework for the proposed study because it directly focuses on vocational interests and educational compatibility. Unlike other personality-based assessments, the RIASEC model specifically measures how student interests align with particular academic programs and career environments. The model classifies individuals into six vocational personality types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional, which makes it highly suitable for course recommendation and career guidance systems. The reviewed studies revealed that students are more likely to achieve academic satisfaction, motivation, persistence, and long-term career success when their educational choices align with their vocational interests. This is the

primary reason why the researchers selected the RIASEC model instead of other psychometric frameworks. The researchers believe that an effective course recommendation system should not only identify personality traits, but should also determine whether the student's interests and abilities are compatible with the nature and demands of a specific academic courses.

The reviewed studies also presented different recommendation systems that utilized artificial intelligence, machine learning, personality analysis, deep learning, and educational profiling technologies to generate personalized recommendations for students. While these systems successfully improved recommendation accuracy and decision-making processes, several limitations were identified. Some studies relied heavily on large and complex datasets, making them difficult to implement in institutions with limited resources. Other systems focused mainly on personality or academic performance while failing to include vocational interests, aptitude testing, and broader psychometric evaluation. In some cases, existing systems emphasized interface design and user interaction rather than intelligent recommendation processes capable of producing comprehensive educational guidance.

These identified limitations created a gap that the present study aims to address. Unlike previous systems, the proposed study integrates admission examination results, RIASEC vocational interest assessment, student profiling, and automated recommendation processes into a single web-based platform specifically intended for incoming students. Through this integrated approach, the proposed system seeks to reduce academic mismatch, improve the efficiency of the admission process, support data-driven academic advising, and provide students with more suitable and personalized course recommendations based on both competence and vocational interest alignment. Ultimately, the study aims to help students make more informed academic decisions while supporting the institution in handling student evaluation and recommendation processes more systematically and efficiently.

CHAPTER 3

RESEACH METHODOLOGY

Introduction

This chapter presents the research methodology used in the development of the Psychometric-Based College Course Recommendation System Using Student Profiling for Tagoloan Community College. It discusses the Software Development Life Cycle (SDLC) model applied in the study, particularly the Agile Flow Model, and explains the different phases involved in the development process. This chapter also presents the methods, tools, procedures, and testing activities conducted by the proponents in designing, developing, evaluating, deploying, and improving the proposed system to ensure that it meets the objectives of the study.

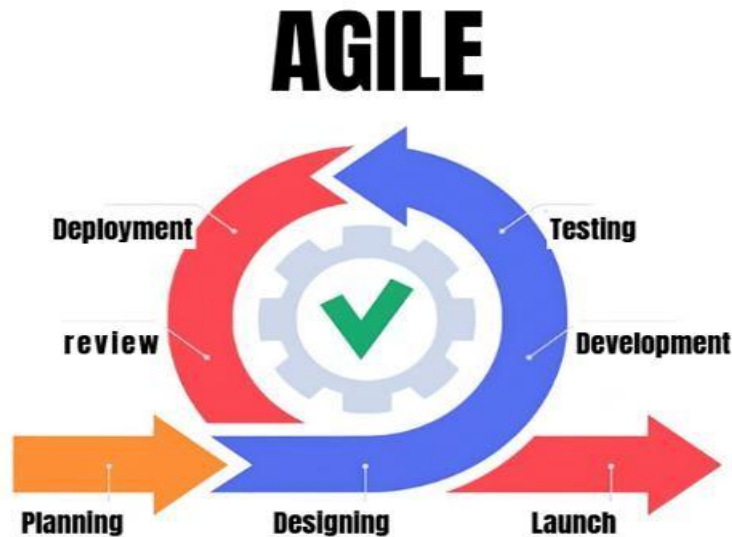


Figure 2.0 Agile SDLC Model – Elfreda, R. P., & Usman, M. L. L. (2026)

Figure 2 presents the Agile Flow Model used in the study. Agile is an iterative and flexible software development approach that focuses on continuous planning, design, development, testing, deployment, review, and launch. Unlike traditional linear development models, Agile allows developers to revise and improve system functionalities based on user

feedback and changing requirements during development. The model consists of the phases of plan, design, develop, test, deploy, review, and launch.

The Agile Flow Model was suitable for the study because the development of the proposed system required continuous evaluation and refinement of the system functionalities, recommendation process, and user interface. Since the system involves psychometric assessment, student profiling, and course recommendation generation, the proponents needed a development approach that allows modifications and improvements throughout the implementation process. The Agile approach enabled the proponents to revise system features and improve system performance based on feedback gathered from the Guidance Office personnel who also serves as a psychometrician.

Through this model, the proponents were able to develop the proposed system in an organized and iterative manner. The Agile Flow Model guided the proponents in continuously designing, developing, testing, evaluating, and improving the system throughout the development process. Furthermore, the model helped ensure the usability, reliability, responsiveness, and effectiveness of the Psychometric-Based College Course Recommendation System Using Student Profiling.

Planning Phase

In this phase, the proponents gathered the necessary information and requirements needed for the development of the proposed system. Interviews were conducted with the Guidance Office personnel, who also serves as a psychometrician, to identify the current problems and needs of the institution regarding course recommendation and student assessment. Through these interviews, the proponents were able to understand the existing admission procedures, student evaluation practices, and challenges encountered in recommending appropriate college courses for incoming students.

The gathered information served as the foundation for identifying the major features and functionalities needed in the system. The proponents identified important system requirements such as student profiling, admission examination processing, RIASEC assessment, course recommendation generation, and report management. Hardware and software requirements necessary for the development of the proposed system were also identified during this phase.

Furthermore, related literature, studies, and existing systems relevant to psychometric assessment, recommendation systems, and student profiling were reviewed to support the development of the study. The information gathered during this phase became the basis for the continuous planning and improvement of the proposed system throughout the development process.

Designing Phase

In this phase, the proponents prepared the overall structure and design of the proposed system based on the gathered requirements. System diagrams, database structures, wireframes,

and user interface prototypes were created to serve as guides before the actual coding and development process began. The design phase focused on organizing the flow of information, system processes, and user interaction within the application.

The proponents developed several diagrams to visualize the system. The Use Case Diagram identified the users and their interactions with the system, while the Data Flow Diagram (DFD) illustrated the movement of student information, examination results, and RIASEC assessment data. The Entity Relationship Diagram (ERD) established the relationships among the database tables including student records, assessment results, and recommended courses. A System Flowchart was also created to show the step-by-step process of data handling, and a System Architecture Diagram was prepared to represent the layered design of the application.

Wireframes and UI prototypes were designed using Figma, which allowed the proponents to visualize the layout, navigation, and overall user experience of the system. These prototypes included student registration forms, assessment interfaces, and recommendation result pages. Tools such as draw.io were used for diagramming, while MySQL Workbench and phpMyAdmin were utilized for database planning. Through the Agile approach, the proponents continuously refined these designs based on evaluation and feedback gathered during development.

Development Phase

In this phase, the proponents developed the Psychometric-Based College Course Recommendation System based on the finalized requirements and system designs. The proponents translated the planned workflows, database structures, and interface designs into a functional webbased application.

The proponents used HTML, CSS, JavaScript, Bootstrap, and Laravel Framework in developing the system interface and functionalities. MySQL and phpMyAdmin were utilized for database management and storage of student records, admission examination scores, RIASEC assessment results, and recommended courses. XAMPP was used as the local server environment during the development process.

The proponents developed the major modules of the system including the User Management Module, Student Profiling Module, RIASEC Assessment Module, Course Recommendation Module, and Report Generation Module. The recommendation process of the system was based on automated evaluation of admission examination scores, student profiles, and RIASEC assessment results to generate appropriate course recommendations for incoming students.

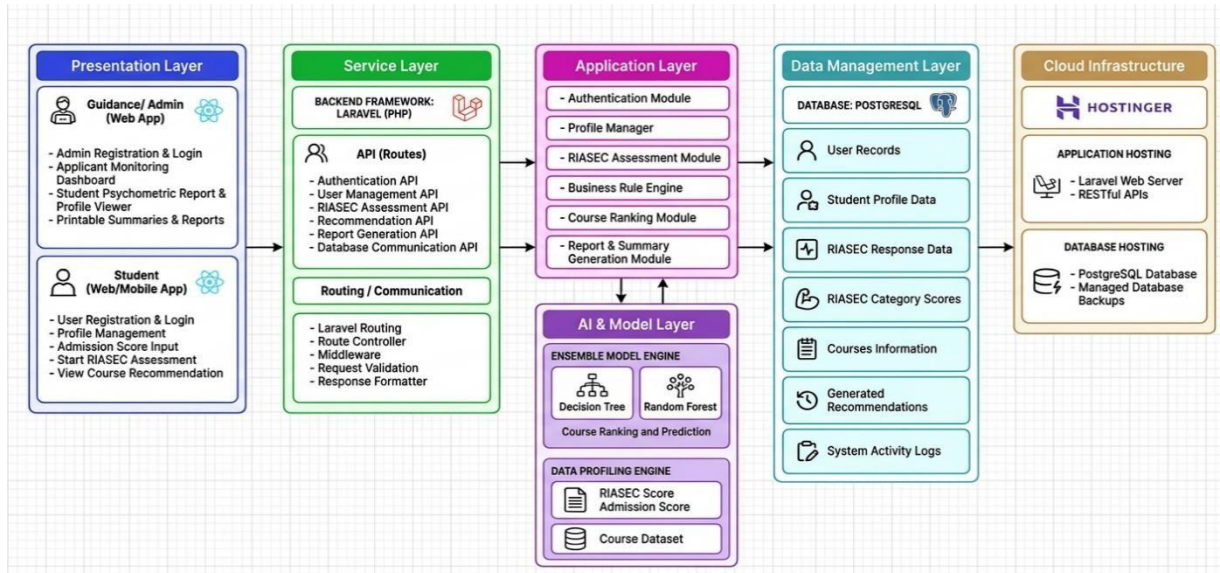


Figure 3.0 System Architectural Design

This architecture represents a layered course recommendation system that uses the RIASEC assessment and machine learning to help students identify suitable college courses. The system begins at the Presentation Layer, where both students and guidance administrators interact with the web or mobile application. Students can register, manage their profiles, input admission scores, take the RIASEC assessment, and view recommended courses, while administrators can monitor applicants, access psychometric reports, and generate printable summaries and reports. The requests from these users are sent to the Service Layer, which uses Laravel (PHP) as the backend framework. This layer manages APIs such as authentication, user management, assessment processing, recommendation generation, and report handling, while also controlling routing, validation, middleware, and communication between components.

The Application Layer serves as the core logic of the system. It contains modules responsible for authentication, profile management, RIASEC assessment processing, business rules, course ranking, and report generation. This layer ensures that student responses and admission data are processed correctly before being analyzed. Connected to it is the AI & Model Layer, where machine learning models such as Decision Tree and Random Forest are used to predict and rank the most suitable courses for students based on their RIASEC scores, admission scores, and course datasets. The results and all related information are then stored in the Data Management Layer using a PostgreSQL database, which keeps records such as user profiles, assessment responses, category scores, recommendations, and system activity logs. Finally, the entire system is deployed in the Cloud Infrastructure Layer using Hostinger, which handles application hosting, RESTful APIs, database management, and backups to ensure the system remains accessible online, secure, and reliable.

Testing Phase

In this phase, the proponents conducted several testing procedures to evaluate the functionality, reliability, usability, and performance of the proposed system. Different test cases and evaluation methods were conducted to ensure that the system was functioning properly and producing accurate results.

Functional testing was conducted to verify whether the major modules and functionalities of the system were operating according to the objectives of the study. The proponents tested system features such as account registration, login authentication, student profiling, admission examination processing, RIASEC assessment scoring, course recommendation generation, and report output.

Integration testing was also conducted to determine whether the different modules and components of the system were properly connected and functioning together as a complete application. This testing ensured that student information, assessment results, and recommendation outputs were processed accurately within the system.

In addition, Scenario-Based Testing (SBT) was applied to simulate real-world usage scenarios. For example, test cases were designed to replicate a student registering, completing the RIASEC assessment, and receiving course recommendations. These scenarios helped validate that the system responded correctly to user actions and produced accurate outputs. Detailed test cases were prepared for each module, specifying inputs, expected outputs, and actual results. This structured approach ensured that errors were identified and corrected systematically.

Usability testing was conducted with the Guidance Office personnel to evaluate accessibility, responsiveness, and ease of navigation. Performance testing was also performed to determine whether the system could efficiently process student information and handle multiple users without errors or delays. Bugs, system errors, and technical issues identified during the testing procedures were corrected and improved to ensure the reliability and effectiveness of the proposed system.

Deployment Phase

In this phase, the proponents prepared and deployed the completed Psychometric-Based College Course Recommendation System for actual use within Tagoloan Community College. Final system configuration, database checking, and module integration were conducted to ensure that all components of the system were functioning properly before deployment.

The proponents organized the final database structure, verified system connectivity, and ensured the accuracy of generated course recommendations. The integration of the student profiling module, RIASEC assessment module, admission examination processing, and administrative functions enabled the system to provide a more organized and systematic process for evaluating incoming students.

The system was deployed using Hostinger Cloud Hosting, which provided a secure and reliable environment for the application. Hostinger handled the web server setup, RESTful API

hosting, database management, and backups. This ensured that the system remained accessible online, with proper scalability and security features. Hosting also allowed administrators and students to access the system remotely, making the admission and guidance process more efficient. The deployment phase further allowed the proponents to gather feedback and identify possible improvements for future system enhancement.

Review Phase

In this phase, the proponents continuously reviewed and evaluated the overall performance and functionality of the proposed system after deployment. Feedback from the Guidance Office personnel who also serves as a psychometrician was gathered to identify possible improvements, technical issues, and user concerns encountered during system usage.

The proponents analyzed the feedback and evaluation results to determine areas that required modification and enhancement. Necessary improvements and adjustments were implemented to improve system usability, reliability, responsiveness, and overall user experience.

Moreover, the review phase supported the continuous improvement process of the Agile Flow Model. Future updates, feature enhancements, and technical improvements may be conducted to further improve the effectiveness and reliability of the Psychometric-Based College Course Recommendation System Using Student Profiling according to the needs of the institution and its users.

Launch Phase

In this final phase, the system was officially launched for institutional use. This stage marked the transition from development to operational application, with ongoing monitoring to ensure effectiveness.

The launch phase ensured that the system was fully integrated into the institution's admission and guidance procedures. It also provided an opportunity to evaluate the system in a real-world environment, ensuring that the functionalities were aligned with institutional needs and user expectations.

Future updates and enhancements will be guided by user feedback and evolving institutional requirements, ensuring that the system remains effective, reliable, and responsive to the needs of Tagoloan Community College.

Reference

Journal

- Al-Dossari, H., Nughaymish, F. A., Al-Qahtani, Z., Alkahlifah, M., & Alqahtani, A. (2024). A machine learning-based career recommendation. *Journal of Trends in Computer Science and Smart Technology*, 6(1), 45–58. <https://irojournals.com/tcsst/article/view/2086>
- Azlan, N. I. D., Awang, M. K., & Mamat, A. R. (2017). Course recommendation based on academic qualification and personality type using rule-based technique. *World Applied Sciences Journal*, 35(10), 2133–2137. World Applied Sciences Journal PDF
- Borgen, F. H., & Betz, N. E. (2022). Career development and decision-making in higher education. *The Career Development Quarterly*. <https://onlinelibrary.wiley.com/journal/21610045>
- Chen, X., & Wang, Y. (2026). Designing a precision career-guidance model based on student psychological profiling in higher education. *Frontiers in Psychology*, 17, 1743896. <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1743896/abstract>
- Christanto, & Singgalen. (2023). Student guidance information system development using the Systems Development Life Cycle (SDLC) with Waterfall model. *Journal of Informatics and Information Systems*. <https://journal-isi.org/index.php/isi/article/view/443/210>
- Commission on Higher Education (CHED) Reports on Student Retention and Course Shifting (2023): (navigate to publications/reports section for retention and course shifting data) retrieved from: <https://ched.gov.ph>
- Dahl, G. B., & Mertens, A. (2020). Academic match and student outcomes in higher education. *Economics of Education Review*. <https://www.sciencedirect.com/journal/economics-ofeducation-review>
- Department of Education. (2025). DepEd Memorandum No. 108, s. 2025: Administration of the Computer-Based National Career Assessment Examination (CB-NCAE). Department of Education, Republic of the Philippines. Retrieved from: https://depedcdo.net/issuance/memoranda/2026?download=8942%3Adm49s2026-administration-of-computer-based-national-career-assessment-examination&utm_source=copilot.com
- Department of Education. (2023). *Basic education report 2023*.

<https://www.deped.gov.ph/2023/01/30/basic-education-report-2023/>

Department of Education. (2025, December 11). *Administration of the computer-based National Career Assessment Examination for school year 2025–2026 (DepEd Memorandum No. 108, s. 2025)*. https://www.deped.gov.ph/wp-content/uploads/DM_s2025_108r.pdf

Education and Employers. (2019). *Motivated to achieve: How encounters with the world of work can influence teenage aspirations and motivation*. <https://www.educationandemployers.org/wp-content/uploads/2019/06/Motivated-toAchieve-Final-Full-report-Embargo-6th-June-1.pdf>

Gardner, W. L., & Martinko, M. J. (1996). Using the Myers-Briggs Type Indicator to study managers: A literature review and research agenda. *Journal of Management*, 22(1), 45–83. <https://doi.org/10.1177/014920639602200103>

Grepon, B. G. S., Baran, N. T., Gumonan, K. M. V. C., Martinez, A. L. M., & Lacs, M. L. E. (2021). Designing and implementing e-school systems: An information systems approach to school management of a community college in Northern Mindanao, Philippines. *arXiv*. <https://arxiv.org/abs/2109.00198>

Hirschi, A. (2020). Career development in the new world of work: Challenges and opportunities. *Frontiers in Psychology*. <https://www.frontiersin.org/articles/10.3389/fpsyg.2020.00002/full>

Holland, J. L. (1997). *Making vocational choices: A theory of vocational personalities and work environments* (3rd ed.). Psychological Assessment Resources. <https://psycnet.apa.org/record/1997-08975-000>

Holland, J. L. (1997). *Making vocational choices: A theory of vocational personalities and work environments* (3rd ed.). Psychological Assessment Resources. <https://www.sciencedirect.com/science/article/abs/pii/S0001879107000371>

Kaya, C., & Karakaya, M. (2021). Interest–ability alignment and academic success. *Education Sciences*. <https://www.mdpi.com/journal/education>

Ogunniran, O. O., & Makinde, O. B. (2022). Decision support systems in education. In M. Fabunmi, G. K. T. Oduro, J. E. Tabotndip, & O. Affreh (Eds.), *Leadership in Education* (pp. 81–99). Education Dialogue Association (EDUDIA) & Institute for Educational Planning and Administration (IEPA). <https://edudia.org/wp-content/uploads/2024/02/1Leadership-in-Education.pdf#page=91>

Organisation for Economic Co-operation and Development. (2019). *PISA 2018 results (Volume II): Where all students can succeed*. OECD Publishing. <https://www.oecd.org/pisa/publications/pisa-2018-results-volume-ii-b5fd1b8f-en.htm>

- Pascua, J. R., & Navalta, J. D. (2023). Student retention practices of Tagoloan Community College. *Global Scientific Journal*, 11(7), 1045–1058. https://www.globalscientificjournal.com/researchpaper/STUDENT_RETENTION_PRACTICES_OF_TAGOLOAN_COMMUNITY_COLLEGE.pdf
- Pascual, N. T. (2014). Factors affecting high school students' career preference: A basis for career planning course. *International Journal of Sciences: Basic and Applied Research*, 16(1), 1–14. <https://gssrr.org/index.php/JournalOfBasicAndApplied/article/view/3008>
- Quintos, C. A., Caballes, D. G., Gapad, E. M., & Valdez, M. R. (2020). Exploring between SHS strand and college course mismatch: Bridging the gap through school policy on intensified career guidance course. *CiiT International Journal of Data Mining and Knowledge Engineering*, 12(10–12), 156–161.
- Quintos, C. A., & Caballes, D. G. (2022). *Exploring between SHS strand and college course mismatch: Bridging the gap through school policy on intensified career guidance course*. <https://www.researchgate.net/publication/361081476>
- Ragunath, K., et al. (2026). Psychometric analysis and activity based assessment system for career guidance. *International Journal of Innovative Research in Science, Engineering and Technology*, 15(4), 112–125. https://www.ijirset.com/upload/2026/april/198_Psychometric%20Analysis%20and%20Activity%20Based%20Assessment%20System%20for%20Career%20Guidance.pdf
- Rahim, A., & Basheer, M. (2025). A hierarchical and multi-tiered personalized career recommender system tailored to individual aptitudes. *Indian Journal of Science and Technology*, 18(2), 89–104. https://www.researchgate.net/publication/388944713_A_Hierarchical_and_Multi-Tiered_Personalized_Career_Recommender_System_Tailored_to_Individual_Aptitudes
- Roy, M., & Sharma, K. (2026). AI-powered career guidance: A scalable model for personalized recommendations using RIASEC and machine learning. *IEEE Access*. https://ieeexplore.ieee.org/document/11373141?fbclid=IwY2xjawR3siVleHRuA2FlbQlXMAbicmlkETFOQlJpTIM0MTR6S2JBcTFYc3J0YwZhchBfaWQQMjIyMDM5MTc4ODIwMDg5MgABHsnNz4RD2qlb35Vu4k_THdA0LAZj646w-MczKSU5pytZg7gLLGws-5dekgSP_aem_e7ezVwEhspyi0IFt3Ig-Vw
- Rudolph, C. W., Lavigne, K. N., & Zacher, H. (2023). Career adaptability and academic outcomes. *Journal of Vocational Behavior*. <https://www.sciencedirect.com/journal/journal-of-vocational-behavior>
- Saputra, M. Y. (2026). *Journal of Instructional Research and Practical Education*. <https://ejournal.papanda.org/index.php/jirpe/article/view/3643>

- Saya, A. (2025). *ERIC document*. <https://eric.ed.gov/?id=EJ1479949>
- Stair, R. M., & Reynolds, G. W. (2020). *Principles of information systems* (13th ed.). Cengage Learning. <https://www.cengage.com/c/principles-of-information-systems-13e-stair/>
- Stajkovic, A. D., Bandura, A., Locke, E. A., Lee, D., & Sergent, K. (2018). Test of three conceptual models of influence of the Big Five personality traits and self-efficacy on academic performance: A meta-analytic path-analysis. *Personality and Individual Differences*, 120, 238–245. <https://doi.org/10.1016/j.paid.2017.08.014>
- Su, R., Stoll, G., & Rounds, J. (2021). Vocational interests and performance: A meta-analysis. *Journal of Vocational Behavior*, 126, 103476. <https://doi.org/10.1016/j.jvb.2020.103476>
- Ugalingan, N. M., & Antonio, E. O. (2025). SRSYS: A web-based solution for efficient school record management. *Aloysian Interdisciplinary Journal of Social Sciences, Education, and Allied Fields*, 1(5), 83–93. <https://zenodo.org/records/15500889>
<https://doi.org/10.3390/su13073936/> <https://www.mdpi.com/2076-3417/11/19/8993>
<https://www.mdpi.com/2079-3200/10/3/43/>
<https://link.springer.com/article/10.1186/s41039-021-00167-7>
- Zhang, W. (2026). Deep learning–driven career analytics model for long-term recommendation systems. *Scientific Reports*, 16, Article 46485. <https://doi.org/10.1038/s41598-026-46485y>

APPENDIX – A

Scanned Copy of the Letter of Request for Interview



TAGOLOAN COMMUNITY COLLEGE
BALUARTE, TAGOLOAN, MISAMIS ORIENTAL
COLLEGE OF INFORMATION TECHNOLOGY



March 23, 2020

Jason Devilleres Ang, RPh, MA, CHRA
Psychometrician, Guidance Designate
Tagoloan Community College (TCC)
Zone 2A Baluarte, Tagoloan

Dear Sir:

Greetings of peace and blessings.

My name is **Rhonald Marc B. Gatchalian**, and I am a 3rd year student at Tagoloan Community College, currently enrolled in the IT Capstone Project 01 course under Bachelor of Science in Information Technology. I am the Project Manager for **Dream. Exe**, working with the following team members:

- Charmie Jane Bagbag
- Emiel Earl Christian B. Virador, Aiza Mae P. Dayondon
- John Wilson C. Comendador
- Jori Lloyd L. Mapalo, Carl Philip N. Ragandang

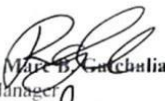
The working title of our Capstone Project is **Development of a Psychometric-Based College Course Recommendation System Using Student Profiling**, and I would like to ask if we may conduct a face-to-face interview with you in your most convenient time.

The purpose of this interview is to understand the organization/business' requirements, expectations, and any specific concerns regarding the project. Specifically, I aim to gather information on:

- Understanding your organization's/business, operations, and current IT usage.
- Identifying any existing challenges, inefficiencies, and pain points in current processes.
- Determining the target users and beneficiaries of the system.
- Exploring possible IT-based solutions that align with your expectations.
- Defining the purpose and impact of the project.
- Clarifying the system's scope, features, constraints, and possible limitations.
- Understanding system workflow, data flow, key transactions, and interactions.
- Paper forms or documents involved and relevant in the process.

Rest assured that confidentiality will be strictly maintained during and after the interview. Should you have any questions please do not hesitate to contact me at gatchalian18rhonaldmare@gmail.com or **099364773957**. I look forward to your positive response with regards to this matter. Thank you and God bless!

Always,


Rhonald Marc B. Gatchalian,
Project Manager

Noted by:

NEPTALE S. ROA III, MIT
Capstone Project 01 Coordinator / Instructor

APPENDIX - B

Scanned Copy of the Interview Questionnaire for the Client



TAGOLOAN COMMUNITY COLLEGE
BALUARTE, TAGOLOAN, MISAMIS ORIENTAL
COLLEGE OF INFORMATION TECHNOLOGY
CPIJRM01 - CAPSTONE PROJECT GUIDE ON WRITING INTERVIEW QUESTIONS



GENERAL DIRECTIONS IN THE CONDUCT OF INTERVIEW TO THE CLIENT

Participation in this interview is voluntary. Client may decline or withdraw at any time. Proponents must obtain explicit consent before recording (audio or video) and explain its purpose. All information will be kept confidential and used only for academic purposes. Client should answer honestly based on their knowledge and experience. Proponents must not share personal or sensitive data beyond authorized members. Client may review their provided answers for accuracy.

ACKNOWLEDGEMENT CLAUSE

I, the undersigned, acknowledge that I have voluntarily agreed to participate in this interview for the IT Capstone Project. I understand that my responses will be kept confidential and used solely for academic purposes. I consent to answer the interview questions to the best of my knowledge and agree to a follow-up interview if necessary.

Client's Full Name:

Jason Devilleres Ang

Signature:

[Handwritten Signature]

Date:

March 24, 2026

Research Questions:

1. How do you describe your experience in administering psychometric assessments for educational or career guidance?
2. What specific types of psychometric tests do you use most often, and why do you prefer them?
3. How do you determine which psychometric assessment is appropriate for a particular student?
4. What is an example of a situation where psychometric assessments influenced a student's course or career path?
5. How do you select or design psychometric instruments for profiling students?
6. How do you conduct the process of administering an aptitude test step by step?
7. How do you ensure the validity and reliability of the psychometric assessments you use?
8. How do you interpret results when multiple assessments are conducted for a single student?
9. What student information do you consider essential for accurate psychometric profiling?
10. How do you collect and verify this information to ensure it accurately reflects the student's abilities and interests?
11. How do demographic or contextual factors influence your decisions in student profiling?
12. How do you integrate qualitative data, such as student interviews and observations, with quantitative test results?
13. How do you analyze psychometric assessment results step by step to recommend suitable courses?
14. What approach do you use when different assessments produce conflicting results for a student?
15. What is an example of a challenging case you encountered, and how did you resolve it using psychometric insights?
16. How do you ensure confidentiality and ethical handling of students' psychometric data?
17. How do you communicate sensitive assessment results to students, especially when the results contradict their expectations?
18. What situation required strong ethical judgment in interpreting or using assessment results?
19. What are the main challenges you encounter when translating psychometric profiling into actionable course recommendations?
20. How would you design a system to automate course recommendations using psychometric data, and what steps, features, or safeguards would you include?

APPENDIX - C

The Riasec Test

THE RIASEC TEST

Follow these easy steps to see where your interests are.

1 Read each statement. If you agree with the statement, fill in the circle. There are no wrong answers!

1. I like to work on cars—	☐						
2. I like to do puzzles—		☐					
3. I am good at working independently—			☐				
4. I like to work in teams—				☐			
5. I am an ambitious person, I set goals for myself—					☐		
6. I like to organize things, (files, desks/offices)—						☐	
7. I like to build things—	☐						
8. I like to read about art and music—			☐				
9. I like to have clear instructions to follow—						☐	
10. I like to try to influence or persuade people—						☐	
11. I like to do experiments—		☐					
12. I like to teach or train people—				☐			
13. I like trying to help people solve their problems—				☐			
14. I like to take care of animals—	☐						
15. I wouldn't mind working 8 hours per day in an office—						☐	
16. I like selling things—						☐	
17. I enjoy creative writing—			☐				
18. I enjoy science—		☐					
19. I am quick to take on new responsibilities—						☐	
20. I am interested in healing people—						☐	
21. I enjoy trying to figure out how things work—	☐						
Total							
R I A S E C							
22. I like putting things together or assembling things—	☐						
23. I am a creative person—				☐			
24. I pay attention to details—						☐	
25. I like to do filing or typing—							☐
26. I like to analyze things (problems/situations)—		☐					
27. I like to play instruments or sing—				☐			
28. I enjoy learning about other cultures—					☐		
29. I would like to start my own business—						☐	
30. I like to cook—		☐					
31. I like acting in plays—				☐			
32. I am a practical person—		☐					
33. I like working with numbers or charts—			☐				
34. I like to get into discussions about issues—					☐		
35. I am good at keeping records of my work—						☐	
36. I like to lead—						☐	
37. I like working outdoors—		☐					
38. I would like to work in an office—						☐	
39. I'm good at math—			☐				
40. I like helping people—						☐	
41. I like to draw—					☐		
42. I like to give speeches—						☐	
Total							
Grand Total							
R I A S E C							

2 Add up the number of filled in circles in each column and then add the two columns together for a grand total.

3 Using your grand total scores from above, transfer the scores for each letter into the appropriate column below.

R	=	Realistic	Total:	_____
I	=	Investigative	Total:	_____
A	=	Artistic	Total:	_____
S	=	Social	Total:	_____
E	=	Enterprising	Total:	_____
C	=	Conventional	Total:	_____

4 Take the three letters with the highest scores and record them under "My Interest Code".

MY INTEREST CODE

5 Turn the page to see what this means!

APPENDIX - C

Results of the Riasec Test

RESULTS OF THE RIASEC TEST

R = Realistic

These people are often good at mechanical or athletic jobs. Good college majors for Realistic people are...

- Agriculture
- Health Assistant
- Computers
- Construction
- Mechanic/Machinist
- Engineering
- Food and Hospitality

Related Pathways

Natural Resources
Health Services
Industrial and Engineering Technology
Arts and Communication

I = Investigative

These people like to watch, learn, analyze and solve problems. Good college majors for Investigative people are...

- Marine Biology
- Engineering
- Chemistry
- Zoology
- Medicine/Surgery
- Consumer Economics
- Psychology

Related Pathways

Health Services
Business
Public and Human Services
Industrial and Engineering Technology

A = Artistic

These people like to work in unstructured situations where they can use their creativity. Good majors for Artistic people are...

- Communications
- Cosmetology
- Fine and Performing Arts
- Photography
- Radio and TV
- Interior Design
- Architecture

Related Pathways

Public and Human Services
Arts and Communication

S = Social

These people like to work with other people, rather than things. Good college majors for Social people are...

- Counseling
- Nursing
- Physical Therapy
- Travel
- Advertising
- Public Relations
- Education

Related Pathways

Health Services
Public and Human Services

E = Enterprising

These people like to work with others and enjoy persuading and performing. Good college majors for Enterprising people are:

- Fashion Merchandising
- Real Estate
- Marketing/Sales
- Law
- Political Science
- International Trade
- Banking/Finance

Related Pathways

Business
Public and Human Services
Arts and Communication

C = Conventional

These people are very detail oriented, organized and like to work with data. Good college majors for Conventional people are...

- Accounting
- Court Reporting
- Insurance
- Administration
- Medical Records
- Banking
- Data Processing

Related Pathways

Health Services
Business
Industrial and Engineering Technology

APPENDIX - C

Career Assessment

Table A1. Career assessment.

1.	I like to work outdoors and with tools and machines. (R)
	Yes
	No
2.	I like to work alone. (R)
	Yes
	No
3.	I enjoy creating original work. (R)
	Yes
	No
4.	I like to be around other people and am interested in how people get along. (I)
	Yes
	No
5.	I am interested in money and/ or politics. (I)
	Yes
	No
6.	I like to work indoors and organize things. (I)
	Yes
	No
7.	I am a practical and straightforward person. (A)
	Yes
	No
8.	I am a curious and discreet person. (A)
	Yes
	No
9.	I am creative and a perfectionist. (A)
	Yes
	No
10.	I am a helpful and caring person. (S)
	Yes
	No
11.	I am talented at managing and willing to take risks. (S)
	Yes
	No
12.	I am a self-disciplined and conservative person. (S)
	Yes
	No
13.	I am good at working with things more than people. (E)
	Yes
	No
14.	I am good at exploring and understanding things or events rather than persuade others. (E)
	Yes
	No
15.	I am good at imagination. (E)
	Yes
	No
16.	I am good at helping, teaching and counselling people more than engaging in mechanical or technical activities. (C)
	Yes
	No
17.	I am good at persuading or directing others more than working on scientific or complicated topics. (C)
	Yes
	No
18.	I am good at following orderly routines and meet clear standards. (C)
	Yes
	No

APPENDIX - D

Client Meeting Documentation

